

ASSIGNMENT NO: 1

UNIVERSITY TRANSPORT & TRACKING SYSTEM



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A. Purpose

The University Transport & Tracking System manages and tracks university-owned transport (buses/shuttles) that ferry students, faculty, and staff between campuses, hostels, and city points. Its core purpose is to provide real-time vehicle location, optimize routes and schedules, and give students visibility into pickup and drop times so they don't waste time waiting. The system operates 8 fixed routes and restricts operation to daytime hours only — late-night routes are not available for safety and operational reasons.

B. Users

- Students — log in with Registration Number and Password; view their own assigned route, live bus location, ETA, and notifications.
- Faculty — view live bus location, ETA, and route information relevant to them.
- Drivers — log in with their own account; start/end trips, report delays or breakdowns.
- Transport Administrators — log in with a separate admin account; manage the 8 routes, schedules, and driver/vehicle assignments.
- Security/Campus Staff — monitor vehicle entry/exit and receive emergency alerts.

C. Stakeholders

- University Administration — owns and funds the system; concerned with cost and operational efficiency.
- Transport Department/Contractors — operate and maintain the vehicles and drivers.
- Students' Parents — interested in safety, particularly given the restriction on late-night routes.
- IT Department — maintains servers, system security, and integration.
- Local Transport Authorities — ensure compliance with traffic and safety regulations.

Roles & Login System

The system supports three distinct login roles, each with its own credentials and access level:

Role	Login Credential	Access / Permissions
Student	Registration Number + Password	View own assigned route, live bus location, ETA, and notifications/alerts.
Admin	Admin Username + Password	Manage all 8 routes, assign drivers/vehicles, add/remove students, send alerts, monitor all vehicles.
Driver	Driver ID + Password	Log in to start/end assigned trip, update trip status, report delays or breakdowns.

D. Functional Requirements

The following functional requirements define the core capabilities of the system, including the login system and the 8-route, daytime-only operating model:

Id	Name	Description	Input	Output	Basic Workflow
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FR-01	Student Login	Allows a registered student to log in using their Registration Number and Password.	Registration Number, Password	Access to student dashboard	Student enters credentials → system validates → dashboard loads with assigned route.
FR-02	Admin Login	Allows an admin to log in with elevated privileges to manage the system.	Admin Username, Password	Access to admin dashboard	Admin enters credentials → system validates role → admin dashboard loads.
FR-03	Live Bus Tracking	Displays real-time GPS location of each bus on a map for the 8 available routes.	GPS coordinates from vehicle device	Live map with bus position	Vehicle device sends location → server updates → map refreshes every 5–10 seconds.
FR-04	ETA Notification	Notifies students of estimated arrival time for their selected route/stop.	Route ID, current bus location	Estimated arrival time	System calculates distance/traffic → generates ETA → displays/pushes to student app.
FR-05	Route & Schedule Management	Allows admin to create, edit, and manage the 8 fixed routes, restricted to daytime hours only (late-night routes unavailable).	Route details, timings, driver/vehicle assignment	Updated route list and schedule	Admin adds/edits route → system validates against allowed time window (e.g., 7:00 AM–8:00 PM) → route published to students.
FR-06	Driver Trip Status	Allows a driver to start/end a trip and report delays or breakdowns.	Trip status, issue report	Updated trip status visible to admin/students	Driver logs in → starts trip → status streams to server → admin/students notified of any delay.
FR-07	Alerts & Notifications	Sends push/SMS alerts to affected students for delays, cancellations, or emergencies.	Event trigger (delay/cancellation/emergency)	Push/SMS notification	Event occurs → system identifies affected route's students → notification dispatched.

E. Non-Functional Requirements

1. Performance: Location updates must refresh at least every 5–10 seconds with under 2-second latency.

2. Availability: The system must be available 99.5% of the time during university operating hours (daytime only).
3. Scalability: Must support tracking of all 8 buses/routes and thousands of concurrent student/faculty app users.
4. Security: All login credentials, location, and personal data must be encrypted in transit and at rest, with strict role-based access control separating Student, Driver, and Admin accounts.
5. Usability: The mobile interface must be simple enough for a first-time student user to log in and find their route without training.

F. Risks

6. GPS/network signal loss in low-connectivity areas leading to inaccurate tracking data.
7. Data privacy breach — leaked login credentials or location data of drivers/students could raise safety and legal concerns.
8. Over-reliance on automation — if the system malfunctions during peak hours with no manual fallback, it could cause large-scale disruption for students relying on the fixed 8-route schedule.

G. Quality Attributes

9. Reliability — the system must consistently provide accurate location and route data, since students plan their day around it.
10. Security — protecting login credentials and location/personal data of students, drivers, and admins is critical.
11. Maintainability — routes, schedules, and driver assignments change often, so admins must be able to update them easily without downtime.

H. Where AI Could Be Used

- Predictive ETA: ML models predicting arrival times based on traffic, weather, and historical patterns.
- Route optimization: AI dynamically adjusts the 8 routes/schedules based on demand patterns (e.g., exam season, holidays).
- Anomaly detection: Flagging unusual vehicle behavior (unexpected stops, route deviation) for safety monitoring.
- Chatbot support: AI assistant answering student queries such as "when's the next bus to Hostel B?"
- Demand forecasting: Predicting peak hours/routes to adjust bus frequency and reduce overcrowding within the daytime operating window.

I. What Could Go Wrong if AI Is Used Incorrectly

- Biased/inaccurate predictions: Incomplete training data (e.g., missing rainy-season patterns) could make ETA predictions consistently wrong, causing students to miss classes.
- Over-automation of safety-critical decisions: If AI reroutes a bus during an emergency without human oversight, it could send a vehicle into an unsafe area or outside the approved daytime window.
- False anomaly alerts: Poorly tuned anomaly detection could flag a routine detour as suspicious, causing unnecessary panic or wasted security response.

- Privacy overreach: Aggressive AI-driven tracking/analytics could monitor student movement beyond what's needed for transport, violating trust — especially sensitive given the Registration Number-linked login.
- Chatbot misinformation: If the AI chatbot confidently gives an incorrect ETA or route, students could miss buses and blame the system.